

Histology

1- Which cell has no nucleus?

- Normoblast
- Erythrocyte
- Basophil
- Neutrophil

2- What is the name of macrophage cells in liver?

- Osteoclast
- Langerhans
- Microglial
- Kupffer

3- Retrograde transportation in the axon is occurred by:

- Kinesin
- Actin
- Dynein
- Talin

4- An exocrine gland that loses cell bodies during secretion, as is the case for the sebaceous gland, is further classified as a.....gland.

- Apocrine
- Holocrine
- Merocrine
- Endocrine

5- In which type of granules in platelets we can find lysosomic enzyme?

- Alpha
- Delta
- Landa
- Beta

6- Which of the following histological structures has Chondroblast?

- Wolkman Canal
- Chondrium Matrix
- Endosteum
- Peroosteom

7- Structure of framework of spleen and lymphnodes are composed of which fibers?

- elastic
- elaunin
- collagen bundles
- reticular

8- What is the following is INCORRECT?

- cardiac muscle and skeletal muscle have sarcomere
- Striation was seen in transverse section of cardiac muscle and skeletal muscle
- Striation was seen in longitudinal section of cardiac muscle and skeletal muscle
- cardiac muscle and smooth muscle nucleus located in center part

9- About Unmyelinated PNS axons, which choice is correct?

- Oligodendrocyte cell starts to envelop multiple axons
- Schwann cell starts to envelop multiple axons
- Schwann cell starts to envelop one axon
- Oligodendrocyte cell starts to envelop one axon

10- Which layer of the epidermis has cells which have keratohyaline granules?

- Stratum basale
- Stratum spinosum
- Stratum corneum
- Stratum granulosum

11- Which of the following is a supporting cell found in the ganglia?

- Satellite cells
- Schwann cells
- Basket cells
- Neuroglia

12- All of these are located in the chemical synapses, except:

- Postsynaptic receptors
- Gap junctions
- Synaptic vesicles
- Mitochondria

13- Which cell has a part in formation of blood brain barrier?

- Astrocyte
- Ependymal cell
- Oligodendrocyte
- Schwann cell

14- Which of the following Is Observed in the histological structure Metaphysis Plate

- Conderoclast
- Chonderocyte
- Osteoclast
- Osteocyte

15- What is line that bisects the dark band in muscle?

- M line
- I band
- A band
- H band

16- What type of muscles have branching or spindle shape cells?

- Smooth- Striated
- Cardiac- Smooth
- Striated - Smooth
- Skeletal- Cardiac

17- A nerve is formed through a collection of

- Glial cells
- Myelin sheet
- Neurons
- Axones

18- Which layer of the epidermis contains cells with no nuclei or organelles?

- spinosum
- corneum
- granulosum
- basale

19- In which cell nucleus has cartwheel or clock face arrangement?

- Plasmacell
- Mastcell
- Fibrocyte
- Fibroblast

20- In which cell granules composed of major basic proteins?

- Basophil
- Lymphocyte
- Neutrophil
- Eosinophil

21- Which cell is the largest white blood cell?

- Neutrophil
- Monocyte
- Basophil
- Lymphocyte

22- Spinal central canal covered by:

- Oligodendrocyte
- Schwann cell
- Microglia
- Ependymal cell

23- Identify the INCORRECT statement

- In pseudostratified epithelia all cells are in contact with the basement membrane.
- The basement membrane consists of two sub layers: the reticular layer and the basal lamina.
- Epithelia are classified by the shape of cell layers
- Microvilli increase the surface area of the epithelia cells

24- In which of the following is Observed Osteogenic cell & Osteoclast ?

- Nothing
- Endosteum
- Periosteum
- Pericardium

25- Which of muscle protein has three subunit(TnT,TnC,TnI)?

- Myosin
- Troponin
- Actin
- Tropomyosin

26- Nissl bodies in the perikaryon of neurons are composed of:

SER and Lysosomes

Golgi and RER

Ribosomes and RER

Ribosomes and Lysosomes

27- In which of the following is Observed only type I collagen ?

Perichondrium

Hyalin Cartilage

Matrix

Elastic Cartilage

28- What do you call the simple squamous epithelium that lines the blood vessels?

Pseudostratified

Epithelioid tissue

Mesothelium

Endothelium

29- What is the covering of an individual muscle fiber?

Sarcolemma

Epimysium

Endomysium

Perimysium

30- In which tissue the nucleus are at the rim of the cells?

Fibroblast

Fibrocyte

Macrophage

Adipocyte

31- Which cell is sensory reception between epidermal cells?

- Merkel
- Lymphocyte
- Melanocyte
- Keratinocyte

32- Which cell secrete antibody in connective tissue?

- Fibrocyte
- Mastcell
- Plasmacell
- Adibocyte

33- What type of glands are the ceruminous glands in ear?

- Apocrine gland
- Eccrine gland
- Sebaceous glands
- Endocrine gland

34- Which types of neurons are located in dorsal root ganglia?

- Pseudounipolar
- Anaxonic
- Bipolar
- Multipolar

35- Which type of junctions is an effective barrier to the diffusion of substances across an epithelium?

- Gap junctions
- Tight junctions
- Hemidesmosomes
- Desmosomes

36- This type of neuron consist of many dendrites and one true axon , and are more:

Multipolar

Unipolar or pseudo unipolar

Bipolar

Apolar

Embryology

37- Implantation of the blastocyst

cannot occur without prior

differentiation of syncytiotrophoblast

.hatching from the zona pellucida

.fragmentation of the primary yolk sac

development of amnioblast

Embryology

38- Major events beginning in the second week of development include:

notochord differentiation

somite formation

angiogenesis

yolk sac development

39- Somites

are the source of cells in the nucleus pulposus

give rise to cardiac muscle

give rise to the cartilages of the branchial arches

are derived from the paraxial mesoderm

40- The outer layer of the umbilical cord is cont

inuuous with, and similar in structure to the

chorionic membrane

amniotic membrane

syncytiotrophoblast

cytotrophoblast

41- Implantation is preceded by an increase in ovarian hCG secretion

usually occurs near the end of the first week following fertilization

requires contact of the zona pellucida with uterine epithelium

cannot occur outside the uterus and uterine tubes

leads to the development of the inner cell mass

42- The primitive streak

is derived from epiblast

first appears in the roof of the amniotic cavity

gives rise to ectomesenchyme of the 1st branchial arch

gives rise to the allantois

43- Which layer of the bilaminar embryonic disc forms part of the lining of the amniotic cavity?

epiblast

hypoblast

blastocyst

trophoblast

44- Pregnancy tests generally are based upon the detection in maternal urine of:

- Estrogen
- Progesterone
- human Chorionic Somatomotropin
- human Chorionic Gonadotrophin

45- The intraembryonic coelom

- it is lined by endoderm
- normally contains only fluid
- becomes the pleural and peritonea
- , but not the pericardial, cavities
- all of the above

46- 21 Mesoderm lateral to the notochord:

- is derived from hypoblast
- differentiates into somite
- migrates cranial to prochordal plate
- induces endoderm differentiation

47- As the follicle grows, the oogonium becomes located in a mound of follicle cells called the:

- vesicular antrum
- oocyte
- cumulus oophorus
- Mural granulosa cells

48- Which of the following is associated with oligohydramnios

- tracheo-esophageal fistula
- . renal agenesis
- myelomeningocele
- anencephal

49- Extraembryonic somatic mesoderm is in close association with:

- syncytiotrophoblast
- cytotrophoblast & amnion
- yolk sac
- syncytio- & cytotrophoblast

50- A 3 year-old male patient presents with epicanthal folds, flat nasal bridge, single palmar crease. The patient is 47 XY on karyotype. What could be the cause of this karyotype?

- Chromosomal mutation
- . Chromosomal inversion
- Chromosomal nondisjunction
- Chromosomal deletion

51- Neural crest

- gives rise to most of the cardiogenic mesoderm
- is derived from the epiblast
- .becomes smooth muscle cells within the wall of the bowel
- .induces development of syncytiotrophoblast

52- The amniotic cavity develops:

- on the tenth day
- within the inner cell mass near the cytotrophoblast
- in extraembryonic mesoderm
- none of the above

53- "Quickening" or the period when fetal movements are commonly first felt by the mother occurs:

- near the end of the first trimester
- around the middle of the second trimester
- at the end of the second trimester
- . at the end of the embryonic period

54- Maternal serum alpha-fetoprotein increases if the fetus has

- oligohydramnios
- Bochdalek hernia
- . Ladd's bands
- anencephaly

55- Myelomeningocele (using clinical terminology)

- is most common at the tip of the coccyx
- leaves the affected individual with paralyzed upper limbs but able to walk
- is caused by a failure of the neural folds to meet and fuse
- is undetectable by the results of amniocentesis

56- A 4-year-old healthy female presents to your office to establish primary care. What phase of meiosis should her oocytes be in?

- Meiosis I - prophase
- Meiosis II - prophase
- Meiosis I - metaphase
- Meiosis II – metaphase

57- 22 Which of the following structures does not turn under onto the ventral surface of the embryo during folding of the head?

- heart
- notochord
- pericardial cavity
- septum transversum

58- The human placenta

- has a maternal component formed by the decidua capsularis
- shows no changes with age
- retains the cytotrophoblast layer of the placental barrier throughout gestation
- has chorionic villi as the structural and functional unit

physiology

59- A motor unit consists of

- An axonal terminal and the group of skeletal muscle fibers it innervates.
- An axonal terminal and whole muscle it innervates.
- A motor neuron and the group of skeletal muscle fibers it innervates.
- A motor neuron and whole muscle it innervates.

60- Which of the following is believed to initiate the contraction process in the skeletal muscle fibers?

- Actin-myosin attachment
- The strong affinity of the troponin C for calcium ions
- Sliding filament mechanism
- Adenosine triphosphatase activity of the myosin head

61- Compared to the skeletal muscle fibers, what is the cause of slow cycling of the myosin cross-bridges in the smooth muscle cells?

- Far less ATPase activity of smooth muscle fibers
- Response to different types of stimuli by smooth muscle fibers
- Higher activity of myosine kinase
- Low affinity to Ca ions in smooth muscle fibers

62- In which of the following muscle, control is exerted mainly by nerve signals?

- Ureters
- Ciliary Muscle
- Bile Duct
- Uterus

63- Which of the following is not matched the excitation contraction coupling in the skeletal muscles?

- Transverse tubules
- Multiple adherent points
- Ryanodine receptors
- Dihydropyridine receptors

64-



What does this figure show?

- Stress relaxation phenomenon
- Smooth muscle contraction
- Latch mechanism
- Smooth muscle relaxation

65- In an electromyography report, findings are compatible with severe axonal lesion of a nerve supply to a skeletal muscle. What symptom do you expect to observe in this patient?

- Hyperplasia of muscle fibers
- Muscle fatigue
- Atrophic changes
- Remodeling of muscle fibers

66- Which of the following is correct about isometric contraction?

- The tension on the muscle remains constant during the contraction.
- The load can not be lifted.
- It occurs when the force of the muscle contraction is greater than the load.
- The muscle shortens.

67- Which of the following best refers to the importance of chemical synapses?

- Synchronous firing of interconnected cells
- Signals amplification
- Rapid firing
- fast conduction velocity

68- Action potentials spread to the interior of the skeletal muscle fiber by way of

- transverse tubules
- terminal cisternae
- longitudinal tubules
- endoplasmic reticulum

69- Which of the following represents one of the electrical synapses properties?

- Presynaptic vesicles
- Synaptic delay
- Chemical transmitter
- Bidirectional transmission

70-  In this figure, what does this sign shows?



Acetylcholine molecules

Acetylcholine estrase

Ca ions

Dens bar

71- Which factor inhibits the active sites of actin filaments?

Inactivation of myosin filaments

ATP binding to actin filaments

Myosin heads detachment

Tropomyosin-troponin complex

72- Which of the following best describes the significant difference between skeletal muscle and large nerve fibers?

Velocity of conduction

Generation of action potential

Voltage gated sodium channels presense

The amount of resting membrane potential

73- Which of the following represents "Acetylcholine gated channels"?



D

B

A

C

74- During rigor mortis, contraction cycle is stopped at step 3 because of

lack of ATP

low affinity of actin filaments to myosin heads

non-functional heads of myosin filaments

ADP and Pi release

75- In this type of muscle, the source of Ca^{2+} ions needed for contraction is just from sarcoplasmic reticulum?

Biceps muscle

Iris muscle of the eye

Blood vessels

Uterus

Histology

سوال 1-

Formation of barrier in CNS and PNS is the function of:

Oligodendrocyte- Schwann cell

Astrocyte- Satellite cell

Oligodendrocyte- Myelin sheat

✓ Astrocyte- Perineurium

سوال 2-

Which type of junctions is for adhesion of cell to basement membrane?

Tight J

Desmosomes

Gap J

✓ Hemidesmosomes

سوال 3-

This type of neuron consist of many dendrites and one true axon , and are more:

Apolar

Bipolar

✓ Multipolar

Unipolar or pseudo unipolar

سوال 4-

Which layer of the epidermis has cells which have keratohyaline granules?

Stratum corneum

Stratum spinosum

✓ Stratum granulosum

Stratum basale

سوال 5- heat production is the most important function in which tissue?

dermis

areolar tissue

✓ brown adipose tissue

tendon

سوال 6-

Axons have all organelles except:

Mitochondria

✓ Ribosomes

Lysosomes

Peroxisome

سوال 7-

Which cell is sensory reception between epidermal cells?

melanocyte

lymphocyte

keratinocyte

✓ merkel

سوال 8-

In which muscle type(s) nucleus located in center part?

smooth muscle and skeletal muscle

smooth muscle and striated muscle

✓ cardiac muscle and smooth muscle

skeletal muscle and cardiac muscle

9- In muscular tissue.....

Muscle cells attached together by junctions

A band include just thin filaments

✓ H band include just thick filaments

M band include just Kratin kinase filaments

10- سوال which one is dense regular connective tissue?

Dermis of skin

Adipose tissue

✓ Tendon

Areolar tissue

11- سوال Which of the following histological structures has Mesenchymal Like cells?

Endosteum

Periosteum

Volkmann Canal

✓ Perichondrium

12- سوال

What type of cellular junctions do osteocyte canals of bone tissue connect to each other?

Desmosome

Adherens

Tight

✓ Gap junction

13- سوال In which of the following is Observed type II collagen ?

Matrix

Perichondrium

Elastic Cartilage

✓ Hyalin Cartilage

14- سوال

Which one makes Nissl bodies?

Smooth Endoplasmic reticulum

Mitochondria

✓ Ribosomes and Rough Endoplasmic reticulum

Synaptic vesicles

15- سوال Which of the following Is Observed in the histological structure

Metaphysis Plate?

Conderoclast

Osteocyte

✓ Chonderocyte

Osteoclast

سوال 16-

What do you call the simple squamous epithelium that lines the serous membranes?

Epithelioid

Pseudostratified

Endothelium

✓ Mesothelium

سوال 17- in which cell we have major basic protein?

macrophage

✓ Eosinophil

lymphocyte

monocyte

سوال 18- structure of framework of spleen and lymphnodes are composed of which fibers?

✓ reticular

elastic

elaunin

collagen bundles

سوال 19-

Spinal central canal lined by:

Schwann cell

Microglia

Oligodendrocyte

✓ Ependymal cell

سوال 20-

Which one is located on the dendrites?

Golgi complex

✓ Dendritic spines

Nucleus

Nissl bodies

سوال 21- In which of the following is Observed Osteoclast ?

Periosteum

Nothing

Pericondrium

✓ Edosteum

سوال 22-

Which cell is a macrophage found in the skin?

- Microglia
- ✓ Langerhans cell
- Dust cell
- Histiocyte

سوال 23-

in which of the cartilage tissue Isnt Observed perichondrium?

- Larynex Cartilage Tissue
- Sulcus Skulla Cartilage Tissue
- ✓ Joint Cartilage Tissue
- Pinna Cartilage Tissue

24- سوال Red blood cell is because it has In the wall.

- ✓ flexible-spectrin
- fragile-actin
- fragile-spectrin
- flexible-fibrinogen

سوال -25

Which type of exocrine glands lose whole of cell bodies during secretion?

Apocrine

endocrine

✓ Holocrine

Merocrine

سوال -26 which cell has fewest number in the normal blood smear?

monocyte

Eosinophil

Neutrophil

✓ Basophil

سوال -27

What is the covering sheath of a muscle?

Sarcolemma

✓ Epimysium

Endomysium

Perimysium

سوال -28

What type of muscle has branching cells?

Striated and Smooth

✓ Cardiac

Skeletal

Smooth

سوال -29

Which type of regulatory protein located in M line of the skeletal muscle?

Neboulin

actinin

✓ Myomesin

Titin

سوال -30

Which of the following Fibers is observed most in the elastic cartilage perichondrium?

Reticular

Elastic

Oxitalan

✓ Collagen I

سوال - 31

Which of the following terms is applied to collection of neuron cell bodies (Soma) in the peripheral nervous system?

White matter

Gray matter

Nucleus

✓ Ganglion

سوال - 32 Russel bodies are characteristic for which cell?

Adipocyte

mastcell

kupfer

✓ plasmacell

سوال - 33

What type of cells makes up the most part of skin hypodermis?

Melanocyte

Fibroblast

✓ Adipocyte

Macrophage

34- following diapedesis only type of leukocytes can return from the tissues back to the blood. Which one is this?

- ✓ Lymphocyte
- basophil
- neutrophil
- monocyte

35- سوال

Identify the INCORRECT statement

- In pseudostratified epithelia all cells are in contact with the basement membrane
- ✓ Epithelia are classified by the shape of cell layers.
- Microvilli increase the surface area of the epithelia cells.
- The basement membrane consists of two sub layers: the reticular layer and the basal lamina.

36-

سوال

About Unmyelinated PNS axons, which choice is correct?

- Oligodendrocyte cell starts to envelop multiple axons
- ✓ Schwann cell starts to envelop multiple axons
- Schwann cell starts to envelop one axon
- Oligodendrocyte cell starts to envelop one axon

Physiology

37- Which transmitter is released from skeletal muscle fibers into the neuro-muscular junction?

- Epinephrine
- ✓ Acetylcholine
- GABA
- Norepinephrine

38- Which of the following is not a type of “muscle remodeling”?

- ✓ Muscle fatigue
- Muscle Atrophy
- Muscle Hypertrophy
- Hyperplasia

39- During discharge of a presynaptic action potential Ca^{2+} enters the presynaptic terminal through at the active zone.

- ✓ Voltage-gated Ca^{2+} channels
- Voltage-gated Na^{+} channels
- Na^{+} - Ca^{2+} channels
- Voltage-gated K^{+} channels

40- Which factor is initiator in the skeletal muscle contraction?

- ✓ Calcium ions
- ATP
- Troponin
- Tropomyosin

41- In which step of contraction cycle, the cycle is stopped during rigor mortis?

- Step 4
- Step 1
- Step 2
- ✓ Step 3

42- Which of the following suggests a rudimentary analog of the transverse tubule system in smooth muscle?

- Synapsin
- ✓ Caveolae
- Troponin complex
- Myosin light chain kinase

43- Which type of receptors produce slower synaptic actions lasting seconds to minutes?

- ✓ Metabotropic receptors
- Ionotropic receptors
- Receptor-channels
- Ligand-gated channels

44- Which of the following maintains "muscle tautness"?

- ✓ Low rate of impulses coming from the spinal cord
- Depletion of muscle glycogen
- Decreased blood flow
- Loss of ATP

45- Which of the following covers active sites in skeletal muscle fibers?

- Myosin filaments
- G actin molecules
- F actin strands
- ✓ Tropomyosine strands

46- Which of the following shows the electrical synapses property?

- Chemical transmitter
- Synaptic delay
- ✓ Bidirectional transmission
- Presynaptic vesicles

47- Which of the following best describes the significant difference between skeletal muscle and large nerve fibers?

- ✓ Velocity of conduction
- Generation of action potential
- Voltage gated sodium channels presense
- Resting membrane potential to an approximately equal extent

48- Which of the following is “regulatory protein” in smooth muscle contraction?

Troponin

Tropomyosin

Actin

✓ Calmodulin

49- What is the importance of chemical synapses?

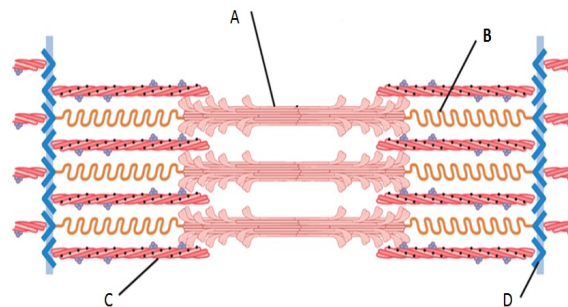
They allow the synchronous firing of interconnected cells

Gap junctions faster the conduction velocity

It is extremely rapid

✓ They can amplify signals.

50- In the figure, which of the following points out “titin protein”?



C

✓ B

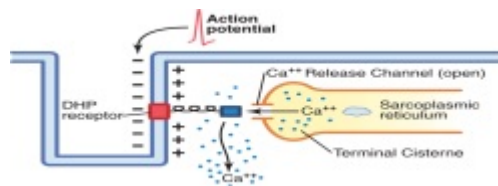
A

D

51- Which of the following is a unitary smooth muscle?

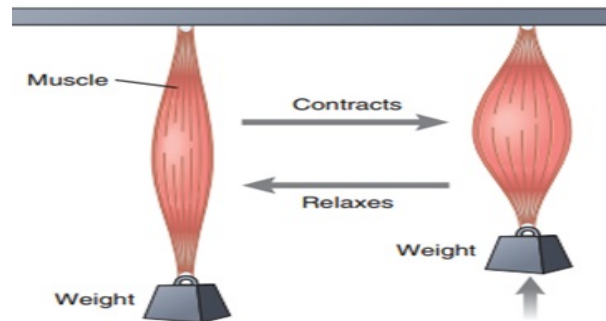
- Ureter
- ✓ Ciliary muscle of the eye
- Gastrointestinal tract
- Bile duct

52- What does this figure show?



- ✓ Excitation-contraction coupling in skeletal muscle
- Excitation- contraction coupling in smooth muscle
- Action potential generation
- Transmitter release in nerve terminal

53- Which phenomenon is shown in this figure?



Isometric contraction



Isotonic contraction

Tetanus

staircase effect

54- Action potentials spread to the interior of the skeletal muscle fiber by way of

longitudinal tubules

reticulum endoplasmic

terminal cisternae



transverse tubules

55- Myosine phosphatase is involved in

- Smooth muscle contraction
- ✓ Smooth muscle relaxation
- Skeletal muscle relaxation
- Skeletal muscle contraction

56- What is the importance of “subneural clefts” in neuro-muscular junction?

- It forms a complex of branching nerve terminals that invaginate into the surface of the muscle fiber.
- It allows the vesicles to move to the active zone of the presynaptic neural membrane.
- ✓ It greatly increases the surface area at which the synaptic transmitter can act.
- It is the space between the nerve terminal and the fiber membrane.

57- In which type of following muscle, the source of ca ions needed for contraction is just from sarcoplasmic reticulum?

- ✓ Soleus muscle
- Iris muscle of the eye
- Uterus
- Blood vessels

58- Which factor causes detachment of the myosin head from the actin filaments in skeletal muscle fibers?.

- ✓ Binding of new ATP
- ATPase activity of the myosin head
- Release of the ADP and phosphate ion that were previously attached to the head
- Tilt of the head toward the arm

Embryology

59- The primitive streak

- ✓ is derived from epiblast
- .first appears in the roof of the amniotic cavity
- gives rise to the allantois
- none of the above

60- Select the correct options below for the structures formed by sclerotome

- ✓ .vertebra
- .dermis
- . limb bud cartilage
- .intervertebral disc

61- Syncytiotrophoblast

- ✓ secretes chorionic gonadotropin
- gives rise to the decidua basalis
- begins to develop on the 4th day post fertilization
- degenerates once cytotrophoblast takes over its function

62- Primordial Germ cells are derived of:

- ✓ .Epiblast
- endodermal cells
- .mesoderm
- hypoblast

63- Which of the following may be associated with polyhydramnios (i.e., may cause polyhydramnios or may result from it)

- duodenal atresia
- anencephaly
- proximal jejunal atresia
- ✓ all of the above

64- In which kind of twin is twin-twin transfusion syndrome the greatest threat to proper development

- ✓ Monochorionic monoamnionic
- Dichorionic monoamnionic
- .Dichorionic diamnionic monozygotic
- Monochorionic diamnionic

65- Human Chorionic Gonadotropin (hCG):

- . is produced by the syncytiotrophoblast
- . enters the maternal blood circulation
- . stimulates the production of progesterone
- ✓ . A, B, and C

66- The zona pellucida plays an important role in

- the formation of the prochordal plate
- neural crest migration
- embryonic blood production
- ✓ preventing premature implantation

67- Phenomena that creates female gametes, oogenesis is ceased at prophase I up till puberty. This describes the DNA content of a female's gametes during her childhood.

- 23 chromosomes, 23 chromatids
- 23 chromosomes, 46 chromatids
- ✓ 46 chromosomes, 92 chromatids
- 46 chromosomes, 46 chromatids

68- Cells in somites give rise to all

- nuclei pulposi
- of the abdominal diaphragm
- bones of the skull
- ✓ skeletal muscles of the trunk

69- The central nervous system forms in the sequence:

- .notochord to neural plate to neural tube
- .neural tube to neural plate to neural groove
- ✓ .neural plate to neural groove to neural tube
- . plate to neural crest to neural zone

70- Maternal serum alpha-fetoprotein increases if the fetus has

- . oligohydramnios
- Bochdalek hernia
- ✓ anencephaly
- .meningocele

71- Which of the following is associated with oligohydramnios

- .tracheo-esophageal fistula
- ✓ renal agenesis
- .myelomeningocele
- .anencephaly

72- Which process in the third week of development converts the embryonic disc into the trilaminar embryo.

- ☐ .placentation
- ☒ .gastrulation
- ☐ .neuralation
- ☐ .somitogenesis

73- In female child at birth , Primordial follicle contain:

- ☒ primary oocyte arrested at prophase I
- ☐ .Secondary oocyte arested at metaphase II
- ☐ .Primordial germ cells
- ☐ .Oogonia

74- Which of the following WILL contribute to the maternal part of the placenta?

- ☐ , chorionic plate
- ☐ , Decidua capsularis
- ☒ decidua basalis
- ☐ , decidua marginalis

75- The correct sequence from cranial to caudal of the secondary brain vesicles is:

- ☐ prosencephalon,mesencephalon,metencephalon,myelencephalon, rhombencephalon
- ☐ .telencephalon,diencephalon, metencephalon, mesencephalon, myelencephalon
- ☒ .telencephalon, diencephalon, mesencephalon, metencephalon, myelencephalon
- ☐ .prosencephalon,diencephalon,mesencephalon,myelencephalon, metencephalon

76- The unspecialized germ cells found in the seminiferous tubules are called

- ☐ .Primary spermatocytes
- ☐ -Secondary spermatocytes
- ☒ -Spermatogonia
- ☐ .Primordial germ cells

77- Which process establishes the 3 definitive germ layers?

- ☐ . Neuralation
- ☒ . Gastrulation
- ☐ . Craniocaudal folding
- ☐ . Lateral folding

78- The average duration of gestation measured from the

first day of the last normal menstrual period

- ☐ 34 weeks
- ☐ .38 weeks
- ☒ .40 weeks
- ☐ .42 weeks

79- During week 2 the blastocyst implants and the outer cell layer in that week becomes:

- ☐ . decidual cells and hypoblast cell layer
- ☒ .cytotrophoblasts and syncytiotrophoblasts
- ☐ .chorion and amniotic membranes
- ☐ .the embryoblast and inner cell mass

80- Between which two layers is the extraembryonic mesoderm located?

- ☐ Epiblast and hypoblast
- ☐ . Syncytiotrophoblast and cytotrophoblast
- ☐ Syncytiotrophoblast and endometrium
- ☒ . Exocoelomic (Heuser's) membrane and cytotrophoblast

81- The intraembryonic coelom forms within:

- ☐ .somites
- ☒ .lateral plate
- ☐ .neural tube
- ☐ .intermediate mesoderm

82- The process of transformation of a spermatid into a spermatozoan is called as

- ☐ . Spermatogenesis
- ☐ .Gametogenesis
- ☒ .Spermiogenesis
- ☐ Sperm maturation

83- Pregnancy tests generally are based upon the detection in maternal urine of:

- . Estrogen
- . Progesterone
- . human Chorionic Somatomotropin
- ✓ . human Chorionic Gonadotrophin

Histology

1- Which of the following is a supporting cell found in the ganglia?

Schwann cells



Satellite cells

Basket cells

Neuroglia

2- What is line that bisects the dark band in muscle?

A band

I band

H band



M line

3- What is the name of macrophage cells in liver?



Kupffer

Langerhans

Microglial

Osteoclast

4- In the histological structure, which of the following is observed Entera Juxtaglomerula Cell ?

Efferent Venule



Glomerule

Glomerula

Afferent Vessel

5- Smallest vessels in body are ?

Venules



Capillaries

Arterioles

Veins

6- Which antibody can cross placenta?

- ☐ IgE
- ☐ IgD
- ☒ IgG
- ☐ IgM

7- An exocrine gland that loses cell bodies during secretion, as is the case for the sebaceous gland, is further classified as a.....gland.

- ☐ Apocrine
- ☐ Merocrine
- ☐ Endocrine
- ☒ Holocrine

8- Spinal central canal covered by:

- ☐ Microglia
- ☒ Ependymal cell
- ☐ Oligodendrocyte
- ☐ Schwann cell

9- Which one is the the thickest layer in the wall of arteries?

- ☒ Media
- ☐ Seroza
- ☐ Intima
- ☐ Adventitia

10- Where is plexus of Meissner in digestive system?

- ☐ Seroza
- ☐ Muscularis
- ☒ Submucosa
- ☐ Mucosa

11- In which type of granules in platelets we can find lysosomic enzyme?

- ☐ Delta
- ☐ Beta
- ☐ Alpha

✓ Landa

12- Which cell has a part in formation of blood brain barrier?

- ✓ Astrocyte
- Oligodendrocyte
- Ependymal cell
- Schwann cell

13- Which layer of the epidermis has cells which have keratohyaline granules?

- Stratum corneum
- Stratum spinosum
- ✓ Stratum granulosum
- Stratum basale

14- In which organ we observe rugae?

- Jejunum
- Ileum
- ✓ Stomach
- Esophagus

15- Nissl bodies in the perikaryon of neurons are composed of:

- ✓ Ribosomes and RER
- Golgi and RER
- Ribosomes and Lysosomes
- SER and Lysosomes

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- Jejunum
- ✓ Ileum
- Colon

17- Discontinuous vessel are the other name of?

Arterioles

- ☐ Venules
- ☐ Continuous capillary
- ☒ Sinusoids

18- In which cell nucleus has cartwheel or clock face arrangement?

- ☐ Fibrocyte
- ☐ Fibroblast
- ☐ Mastcell
- ☒ Plasmacell

19- All of these are located in the chemical synapses, except:

- ☐ Synaptic vesicles
- ☒ Gap junctions
- ☐ Mitochondria
- ☐ Postsynaptic receptors

20- Which of muscle protein has three subunit(TnT,TnC,TnI)?

- ☒ Troponin
- ☐ Myosin
- ☐ Tropomyosin
- ☐ Actin

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- ☒ Perichondrium
- ☐ Wolkman Canal
- ☐ Endosteum
- ☐ Periosteum

22- Which of the following Is Observed in the histological structure Metaphysis Plate?

- ☐ Chonderogenesis Cell
- ☐ Osteocyte
- ☐ Osteogenic Cell
- ☒ Prolifration Chonderoblast

23- In which cell granules composed of major basic proteins?



- Eosinophil
- Neutrophil
- Basophil
- Lymphocyte

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- Kinesin
- Talin
- Actin

25- Which cell secrete antibody in connective tissue?



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- Plasmacell
- Fibrocyte
- Adibocyte

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- Ileum
- Colon
- Jejunum

27- Which layer of the epidermis contains cells with no nuclei or organelles?



- corneum
- spinosum
- granulosum
- basale

28- Which cell is the largest white blood cell?

- Lymphocyte
- Basophil
- Monocyte



Neutrophil

29- Identify the INCORRECT statement

In pseudostratified epithelia all cells are in contact with the basement membrane.

Microvilli increase the surface area of the epithelia cells



Epithelia are classified by the shape of cell layers

The basement membrane consists of two sub layers: the reticular layer and the basal lamina.

30- In which tissue the nucleus are at the rim of the cells?

Macrophage



Adipocyte

Fibroblast

Fibrocyte

31- Which cell is sensory reception between epidermal cells?

Melanocyte

Keratinocyte

Lymphocyte



Merkel

32- Which cell has no nucleus?

Normoblast

Neutrophil



Erythrocyte

Basophil

33- In the histological structure, which of the following is observed Macuka Densa epithelial cells?

Henle

Collect Tubule

PCT



DCT

34- What is the following is INCORRECT?

Striation was seen in longitudinal section of cardiac muscle and skeletal muscle
cardiac muscle and smooth muscle nucleus located in center part



Striation was seen in transverse section of cardiac muscle and skeletal muscle
cardiac muscle and skeletal muscle nucleus have sarcomere

35- A nerve is formed through a collection of

Glial cells



Axones

Myelin sheet

Neurons

36- Which types of neurons are located in dorsal root ganglia?



Pseudounipolar

Anaxonic

Bipolar

Multipolar

37- Which of the following Is Observed in the histological structure, Theracea Tissue?

Discontinuous Ringle Form Hyalin cartilage tissue



Continuous Ringle Form Hyalin cartilage tissue

Continuous Ringle Form elastic cartilage tissue

Discontinuous Ringle Form elastic cartilage tissue

38- What do you call the simple squamous epithelium that lines the blood vessels?

Epithelioid tissue

Pseudostratified



Endothelium

Mesothelium

39- سوال Which of the following is Observed in the histological structure of the Alveolar Lung Tissue?

Cell Clara

Pneumocytes VI



Pneumocytes I

Cuboiedal simple cell

40- What type of glands are the ceruminous glands in ear?

- ✓ Apocrine gland
- Endocrine gland
- Eccrine gland
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41- Which type of junctions is an effective barrier to the diffusion of substances across an epithelium?

- Hemidesmosomes
- Desmosomes
- Gap junctions
- ✓ Tight junctions

42- What is the covering of an individual muscle fiber?

- Epimysium
- Perimysium
- Sarcolemma
- ✓ Endomysium

Anatomy

43- belong to the large intestine.

- Ileum
- ✓ Ascending colon
- Appendix
- Duodenum

44- Which one is a short bone in our body?

- Patella
- ✓ A carpal bone
- Vertebral column
- Sesamoid

45- In pronation of hand which one can be true?

- Arm look upward
- Shoulder look forward
- Palm look upward
- ✓ Palm look downward

46- Shoulder girdle made by

- Pelvic and femur
- Sternum and hip
- Pubic, ischium and ileum
- ✓ Clavicle and scapula

47- Which one is true?

- Supination means turn right
- Medial means near to the out line
- Upper means close to the horizontal plane
- ✓ Distal means far from the root

48- which one is not true?

- The umbilical is one of the important landmark in abdomen
- Portal vein collect bloods of the digestive system
- You can palpate Iliac crest over the skin
- ✓ Portal vein made by kidney and cava veins

49- Which one is not intra peritoneum organ?

- spleen
- ✓ kidneys
- Liver
- appendix

50- Which one is False?

- False pelvic contain the most lower part of the small intestine
- ✓ True pelvic is a space above the pelvic inlet

True pelvic contain rectum

False pelvic is a space above the pelvic inlet

51- Which sentences is true?

There are 2 muscles at the side of the abdomen

Muscles at pectoral attached to scapula

Muscles at medial side of elbow are extensor

✓ There is lordosis in the vertebral column at the lumbar part

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✓ T8

T10

L1

T12

53- Which one covers by the diaphragm?

kidneys

✓ liver

pancreases

Most lower part of the stomach

54- Which one belong to lesser finger side of hand?

Extensor muscles

Flexor muscles

Pectoral muscles

✓ Hypothenar muscles

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A frontal axis

✓ A vertical axis

A sagittal axis

A transverse axis

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Abduction is thigh is the main treatment for CDH

In male child at birth time testes descending is completed

Dislocation of hip (CDH) is important disease in female child

✓ In female child it takes time to descending the ovaries after birth

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✓ Tibia and fibula

Fibula and radius

Radius and tibia

Ulna and radius

58- Which one located in left hypochondriac region?

✓ Spleen

Liver

Rectum

Appendix process

59- Which one is not true?

✓ The sigmoid has 3 tenia coli

Stomach and esophagus each one has 10 inches' length

Large intestine is lobulated by tenia coli

The fundus of the stomach always fills by air

60- lesser sac is

The police of the abdomen

A cavity inside of the stomach

✓ A space behind the stomach

A space beyond the anterior wall of abdomen

61- A horizontal plane divide body into parts.

Anterior and posterior

Outer and inner

✓ Superior and inferior

Medial and lateral

62- Which sentences is true?

Patella is a short bone that articulate to femur



ribs belong to flat bone category

Tibia and fibula both take part to making knee joint

Femur articulate by radius

Embryology

63- Implantation of the blastocyst

cannot occur without prior

differentiation of syncytiotrophoblast



.hatching from the zona pellucida

.fragmentation of the primary yolk sac

development of amnioblast

Embryology

64- Implantation is preceded by an increase in ovarian hCG secretion



usually occurs near the end of the first week following fertilization

requires contact of the zona pellucida with uterine epithelium

cannot occur outside the uterus and uterine tubes

leads to the development of the inner cell mass

65- The intraembryonic coelom



it is lined by endoderm

normally contains only fluid

becomes the pleural and peritonea

, but not the pericardial, cavities

all of the above

66- A 3 year-old male patient presents with epicanthal folds, flat nasal bridge, single palmar crease. The patient is 47 XY on karyotype. What could be the cause of this karyotype?

Chromosomal mutation

. Chromosomal inversion



Chromosomal nondisjunction

Chromosomal deletion

67- Major events beginning in the second week of development include:

notochord differentiation

somite formation

angiogenesis



yolk sac development

68- The primitive streak



is derived from epiblast

first appears in the roof of the amniotic cavity

gives rise to ectomesenchyme of the 1st branchial arch

gives rise to the allantois

69- The outer layer of the umbilical cord is cont

inuous with, and similar in structure to the

chorionic membrane



amniotic membrane

syncytiotrophoblast

cytotrophoblast

70- Which of the following is associated with oligohydramnios

tracheo-esophageal fistula



. renal agenesis

myelomeningocele

anencephal

71- Which layer of the bilaminar embryonic disc forms part of the lining of the amniotic cavity?

epiblast



hypoblast

blastocyst

trophoblast

72- Maternal serum alpha-fetoprotein increases if the fetus has

oligohydramnios

Bochdalek hernia

. Ladd's bands



anencephaly

73- Myelomeningocele (using clinical terminology)

is most common at the tip of the coccyx



leaves the affected individual with paralyzed upper limbs but able to walk

is caused by a failure of the neural folds to meet and fuse

is undetectable by the results of amniocentesis

74- As the follicle grows, the oogonium becomes located in a mound of follicle cells called the:

vesicular antrum

oocyte



cumulus oophorus

Mural granulosa cells

75- Somites

are the source of cells in the nucleus pulposus

give rise to cardiac muscle

give rise to the cartilages of the branchial arches



are derived from the paraxial mesoderm

76- Extraembryonic somatic mesoderm is in close association with:

- ☐ syncytiotrophoblast
- ☒ cytotrophoblast & amnion
- ☐ yolk sac
- ☐ syncytio- & cytotrophoblast

77- 21 Mesoderm lateral to the notochord:

- ☐ is derived from hypoblast
- ☒ differentiates into somite
- ☐ migrates cranial to prochordal plate
- ☐ induces endoderm differentiation

78- "Quickening" or the period when fetal movements are commonly first felt by the mother occurs:

- ☐ near the end of the first trimester
- ☒ around the middle of the second trimester
- ☐ at the end of the second trimester
- ☐ . at the end of the embryonic period

79- The amniotic cavity develops:

- ☐ on the tenth day
- ☒ within the inner cell mass near the cytotrophoblast
- ☐ in extraembryonic mesoderm
- ☐ none of the above

80- 22 Which of the following structures does not turn under onto the ventral surface of the embryo during folding of the head?

- ☐ heart
- ☒ notochord
- ☐ pericardial cavity
- ☐ septum transversum

81- Neural crest

- gives rise to most of the cardiogenic mesoderm
- ✓ is derived from the epiblast
- .becomes smooth muscle cells within the wall of the bowel
- .induces development of syncytiotrophoblast

82- The human placenta

- has a maternal component formed by the decidua capsularis
- shows no changes with age
- retains the cytotrophoblast layer of the placental barrier throughout gestation
- ✓ has chorionic villi as the structural and functional unit

83- Pregnancy tests generally are based upon the detection in maternal urine of:

- Estrogen
- Progesterone
- human Chorionic Somatommotropin
- ✓ human Chorionic Gonadotrophin

84- A 4-year-old healthy female presents to your office to establish primary care. What phase of meiosis should her oocytes be in?

- ✓ Meiosis I - prophase
- Meiosis II - prophase
- Meiosis I - metaphase
- Meiosis II – metaphase

Histology

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Fibrocyte

Fibroblast

✓ Adipocyte

Macrophage

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Ribosomes and RER

Golgi and RER

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Most lower part of the stomach

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Liver

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✓ A space behind the stomach

A space beyond the anterior wall of abdomen

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True pelvic contain rectum

✓ True pelvic is a space above the pelvic inlet

False pelvic contain the most lower part of the small intestine

51- belong to the large intestine.

Ileum

Appendix

✓ Ascending colon

Duodenum

52- Which sentence is INCORRECT?

In the cerebellum white matter located in center

In the spinal cord gray matter located in center.

In the cerebrum white matter located in center.

✓ In the spinal cord white matter located in center

53- Which one is a short bone in our body?

Vertebral column

Sesamoid

✓ A carpal bone

Patella

54- Which one belong to lesser finger side of hand?

Extensor muscles

Flexor muscles

Pectoral muscles

✓ Hypothenar muscles

55- These two bones make the leg bony structure.

Ulna and radius

✓ Tibia and fibula

Radius and tibia

Fibula and radius

56- Rotation happens in?

- A frontal axis
- A sagittal axis
- A transverse axis
- ✓ A vertical axis

57- Which part ISNOT apart of brain stem ?

- Pons
- Midbrain
- Medulla
- ✓ Cerebellum

58- Which one is not true?

- Stomach and esophagus each one has 10 inches' length
- Large intestine is lobulated by tenia coli
- ✓ The sigmoid has 3 tenia coli
- The fundus of the stomach always fills by air

59- Which one located in left hypochondriac region?

- Rectum
- ✓ Spleen
- Appendix process
- Liver

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- Arm look upward
- Shoulder look forward
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Muscles at medial side of elbow are extensor

There are 2 muscles at the side of the abdomen

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Pelvic and femur

Pubic, ischium and ileum

✓ Clavicle and scapula

Sternum and hip

63- Inferior cava cross the diaphragm at

L1

✓ T8

T10

T12

64- Regarding the cranial nerves

The facial nerve is purely motor

The trigeminal nerve is purely sensory

✓ The abducens nerve is purely motor

The trochlear nerve is purely sensory

65- A horizontal plane divide body into parts.

Medial and lateral

✓ Superior and inferior

Anterior and posterior

Outer and inner

آزمون: تشریح و بافت مدت آزمون: 66 بر اساس دفترچه کاربر

66- Which sulcus on the cerebral cortex located between temporal and frontal lobes?

Central sulcus

Lunate sulcus



Lateral sulcus

Sylvian sulcus

Histology

1- سوال

Formation of barrier in CNS and PNS is the function of:

Oligodendrocyte- Schwann cell

Astrocyte- Satellite cell

Oligodendrocyte- Myelin sheat

✓ Astrocyte- Perineurium

2- سوال

Which type of junctions is for adhesion of cell to basement membrane?

Tight J

Desmosomes

Gap J

✓ Hemidesmosomes

3- سوال

This type of neuron consist of many dendrites and one true axon , and are more:

Apolar

Bipolar

✓ Multipolar

Unipolar or pseudo unipolar

سوال 4-

Which layer of the epidermis has cells which have keratohyaline granules?

Stratum corneum

Stratum spinosum

✓ Stratum granulosum

Stratum basale

سوال 5- heat production is the most important function in which tissue?

dermis

areolar tissue

✓ brown adipose tissue

tendon

سوال 6-

Axons have all organelles except:

Mitochondria

✓ Ribosomes

Lysosomes

Peroxisome

سوال 7-

Which cell is sensory reception between epidermal cells?

melanocyte

lymphocyte

keratinocyte



merkel

سوال 8-

In which muscle type(s) nucleus located in center part?

smooth muscle and skeletal muscle

smooth muscle and striated muscle



cardiac muscle and smooth muscle

skeletal muscle and cardiac muscle

9- In muscular tissue.....

Muscle cells attached together by junctions

A band include just thin filaments



H band include just thick filaments

M band include just Kratin kinase filaments

10- سوال which one is dense regular connective tissue?

Dermis of skin

Adipose tissue

✓ Tendon

Areolar tissue

11- سوال Which of the following histological structures has Mesenchymal Like cells?

Endosteum

Periosteum

Volkmann Canal

✓ Perichondrium

12- سوال

What type of cellular junctions do osteocyte canals of bone tissue connect to each other?

Desmosome

Adherens

Tight

✓ Gap junction

13- سوال In which of the following is Observed type II collagen ?

Matrix

Perichondrium

Elastic Cartilage

✓ Hyalin Cartilage

14- سوال

Which one makes Nissl bodies?

Smooth Endoplasmic reticulum

Mitochondria

✓ Ribosomes and Rough Endoplasmic reticulum

Synaptic vesicles

15- سوال Which of the following Is Observed in the histological structure

Metaphysis Plate?

Conderoclast

Osteocyte

✓ Chonderocyte

Osteoclast

سوال 16-

What do you call the simple squamous epithelium that lines the serous membranes?

- Epithelioid
- Pseudostratified
- Endothelium
- ✓ Mesothelium

سوال 17- in which cell we have major basic protein?

- macrophage
- ✓ Eosinophil
- lymphocyte
- monocyte

سوال 18- structure of framework of spleen and lymphnodes are composed of which fibers?

- ✓ reticular
- elastic
- elaunin
- collagen bundles

سوال 19-

Spinal central canal lined by:

Schwann cell

Microglia

Oligodendrocyte

✓ Ependymal cell

سوال 20-

Which one is located on the dendrites?

Golgi complex

✓ Dendritic spines

Nucleus

Nissl bodies

سوال 21- In which of the following is Observed Osteoclast ?

Periosteum

Nothing

Pericondrium

✓ Edosteum

سوال 22-

Which cell is a macrophage found in the skin?

- Microglia
- ✓ Langerhans cell
- Dust cell
- Histiocyte

سوال 23-

in which of the cartilage tissue Isnt Observed perichondrium?

- Larynex Cartilage Tissue
- Sulcus Skull Cartilage Tissue
- ✓ Joint Cartilage Tissue
- Pinna Cartilage Tissue

24- سوال Red blood cell is because it has In the wall.

- ✓ flexible-spectrin
- fragile-actin
- fragile-spectrin
- flexible-fibrinogen

سوال -25

Which type of exocrine glands lose whole of cell bodies during secretion?

Apocrine

endocrine

✓ Holocrine

Merocrine

سوال -26 which cell has fewest number in the normal blood smear?

monocyte

Eosinophil

Neutrophil

✓ Basophil

سوال -27

What is the covering sheath of a muscle?

Sarcolemma

✓ Epimysium

Endomysium

Perimysium

سوال -28

What type of muscle has branching cells?

Striated and Smooth



Cardiac

Skeletal

Smooth

سوال -29

Which type of regulatory protein located in M line of the skeletal muscle?

Neboulin

actinin



Myomesin

Titin

سوال -30

Which of the following Fibers is observed most in the elastic cartilage perichondrium?

Reticular

Elastic

Oxitalan



Collagen I

سوال - 31

Which of the following terms is applied to collection of neuron cell bodies (Soma) in the peripheral nervous system?

White matter

Gray matter

Nucleus

✓ Ganglion

سوال - 32 Russel bodies are characteristic for which cell?

Adipocyte

mastcell

kupfer

✓ plasmacell

سوال - 33

What type of cells makes up the most part of skin hypodermis?

Melanocyte

Fibroblast

✓ Adipocyte

Macrophage

34- سوال following diapedesis only type of leukocytes can return from the tissues back to the blood. Which one is this?

- ✓ Lymphocyte
- basophil
- neutrophil
- monocyte

35- سوال

Identify the INCORRECT statement

- In pseudostratified epithelia all cells are in contact with the basement membrane
- ✓ Epithelia are classified by the shape of cell layers.
- Microvilli increase the surface area of the epithelia cells.
- The basement membrane consists of two sub layers: the reticular layer and the basal lamina.

36-

سوال

About Unmyelinated PNS axons, which choice is correct?

- Oligodendrocyte cell starts to envelop multiple axons
- ✓ Schwann cell starts to envelop multiple axons
- Schwann cell starts to envelop one axon
- Oligodendrocyte cell starts to envelop one axon

Physiology

37- Which transmitter is released from skeletal muscle fibers into the neuro-muscular junction?

- Epinephrine
- ✓ Acetylcholine
- GABA
- Norepinephrine

38- Which of the following is not a type of “muscle remodeling”?

- ✓ Muscle fatigue
- Muscle Atrophy
- Muscle Hypertrophy
- Hyperplasia

39- During discharge of a presynaptic action potential Ca^{2+} enters the presynaptic terminal through at the active zone.

- ✓ Voltage-gated Ca^{2+} channels
- Voltage-gated Na^{+} channels
- Na^{+} - Ca^{2+} channels
- Voltage-gated K^{+} channels

40- Which factor is initiator in the skeletal muscle contraction?

- ✓ Calcium ions
- ATP
- Troponin
- Tropomyosin

41- In which step of contraction cycle, the cycle is stopped during rigor mortis?

Step 4

Step 1

Step 2

✓ Step 3

42- Which of the following suggests a rudimentary analog of the transverse tubule system in smooth muscle?

Synapsin

✓ Caveolae

Troponin complex

Myosin light chain kinase

43- Which type of receptors produce slower synaptic actions lasting seconds to minutes?

✓ Metabotropic receptors

Ionotropic receptors

Receptor-channels

Ligand-gated channels

44- Which of the following maintains "muscle tautness"?

✓ Low rate of impulses coming from the spinal cord

Depletion of muscle glycogen

Decreased blood flow

Loss of ATP

45- Which of the following covers active sites in skeletal muscle fibers?

- Myosin filaments
- G actin molecules
- F actin strands
- ✓ Tropomyosine strands

46- Which of the following shows the electrical synapses property?

- Chemical transmitter
- Synaptic delay
- ✓ Bidirectional transmission
- Presynaptic vesicles

47- Which of the following best describes the significant difference between skeletal muscle and large nerve fibers?

- ✓ Velocity of conduction
- Generation of action potential
- Voltage gated sodium channels presense
- Resting membrane potential to an approximately equal extent

48- Which of the following is “regulatory protein” in smooth muscle contraction?

Troponin

Tropomyosin

Actin

✓ Calmodulin

49- What is the importance of chemical synapses?

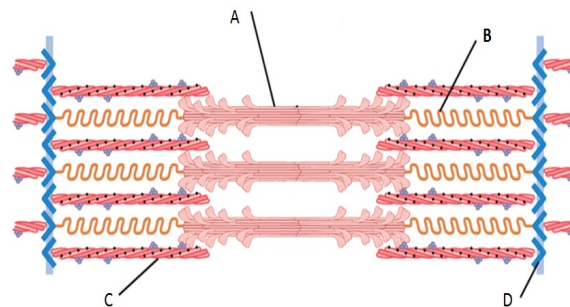
They allow the synchronous firing of interconnected cells

Gap junctions faster the conduction velocity

It is extremely rapid

✓ They can amplify signals.

50- In the figure, which of the following points out “titin protein”?



C

✓ B

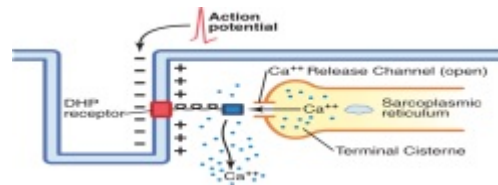
A

D

51- Which of the following is a unitary smooth muscle?

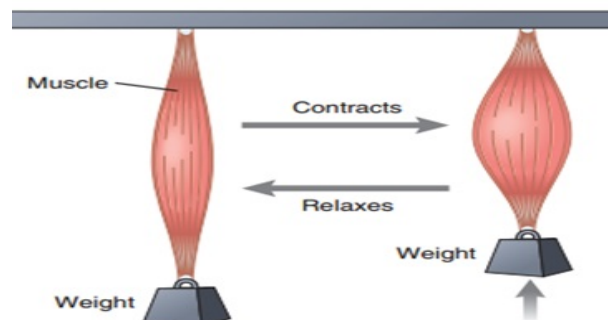
- Ureter
- ✓ Ciliary muscle of the eye
- Gastrointestinal tract
- Bile duct

52- What does this figure show?



- ✓ Excitation-contraction coupling in skeletal muscle
- Excitation- contraction coupling in smooth muscle
- Action potential generation
- Transmitter release in nerve terminal

53- Which phenomenon is shown in this figure?



Isometric contraction



Isotonic contraction

Tetanus

staircase effect

54- Action potentials spread to the interior of the skeletal muscle fiber by way of

longitudinal tubules

reticulum endoplasmic

terminal cisternae



transverse tubules

55- Myosine phosphatase is involved in

- ☐ Smooth muscle contraction
- ☒ Smooth muscle relaxation
- ☐ Skeletal muscle relaxation
- ☐ Skeletal muscle contraction

56- What is the importance of “subneural clefts” in neuro-muscular junction?

- ☐ It forms a complex of branching nerve terminals that invaginate into the surface of the muscle fiber.
- ☐ It allows the vesicles to move to the active zone of the presynaptic neural membrane.
- ☒ It greatly increases the surface area at which the synaptic transmitter can act.
- ☐ It is the space between the nerve terminal and the fiber membrane.

57- In which type of following muscle, the source of ca ions needed for contraction is just from sarcoplasmic reticulum?

- ☒ Soleus muscle
- ☐ Iris muscle of the eye
- ☐ Uterus
- ☐ Blood vessels

58- Which factor causes detachment of the myosin head from the actin filaments in skeletal muscle fibers?.

- ☒ Binding of new ATP
- ☐ ATPase activity of the myosin head
- ☐ Release of the ADP and phosphate ion that were previously attached to the head
- ☐ Tilt of the head toward the arm

Embryology

59- The primitive streak

- ✓ is derived from epiblast
- .first appears in the roof of the amniotic cavity
- gives rise to the allantois
- none of the above

60- Select the correct options below for the structures formed by sclerotome

- ✓ .vertebra
- .dermis
- . limb bud cartilage
- .intervertebral disc

61- Syncytiotrophoblast

- ✓ secretes chorionic gonadotropin
- gives rise to the decidua basalis
- begins to develop on the 4th day post fertilization
- degenerates once cytotrophoblast takes over its function

62- Primordial Germ cells are derived of:

- ✓ .Epiblast
- endodermal cells
- .mesoderm
- hypoblast

63- Which of the following may be associated with polyhydramnios (i.e., may cause polyhydramnios or may result from it)

- duodenal atresia
- anencephaly
- proximal jejunal atresia
- ✓ all of the above

64- In which kind of twin is twin-twin transfusion syndrome the greatest threat to proper development

- ✓ Monochorionic monoamnionic
- Dichorionic monoamnionic
- .Dichorionic diamnionic monozygotic
- Monochorionic diamnionic

65- Human Chorionic Gonadotropin (hCG):

- . is produced by the syncytiotrophoblast
- . enters the maternal blood circulation
- . stimulates the production of progesterone
- ✓ . A, B, and C

66- The zona pellucida plays an important role in

- the formation of the prochordal plate
- neural crest migration
- embryonic blood production
- ✓ preventing premature implantation

67- Phenomena that creates female gametes, oogenesis is ceased at prophase I up till puberty. This describes the DNA content of a female's gametes during her childhood.

- 23 chromosomes, 23 chromatids
- 23 chromosomes, 46 chromatids
- ✓ 46 chromosomes, 92 chromatids
- 46 chromosomes, 46 chromatids

68- Cells in somites give rise to all

- nuclei pulposi
- of the abdominal diaphragm
- bones of the skull
- ✓ skeletal muscles of the trunk

69- The central nervous system forms in the sequence:

- .notochord to neural plate to neural tube
- .neural tube to neural plate to neural groove
- ✓ .neural plate to neural groove to neural tube
- . plate to neural crest to neural zone

70- Maternal serum alpha-fetoprotein increases if the fetus has

- . oligohydramnios
- Bochdalek hernia
- ✓ anencephaly
- .meningocele

71- Which of the following is associated with oligohydramnios

- .tracheo-esophageal fistula
- ✓ renal agenesis
- .myelomeningocele
- .anencephaly

72- Which process in the third week of development converts the embryonic disc into the trilaminar embryo.

- ☐ .placentation
- ☒ .gastrulation
- ☐ .neuralation
- ☐ .somitogenesis

73- In female child at birth , Primordial follicle contain:

- ☒ primary oocyte arrested at prophase I
- ☐ .Secondary oocyte arested at metaphase II
- ☐ .Primordial germ cells
- ☐ .Oogonia

74- Which of the following WILL contribute to the maternal part of the placenta?

- ☐ , chorionic plate
- ☐ , Decidua capsularis
- ☒ decidua basalis
- ☐ , decidua marginalis

75- The correct sequence from cranial to caudal of the secondary brain vesicles is:

- ☐ prosencephalon,mesencephalon,metencephalon,myelencephalon, rhombencephalon
- ☐ .telencephalon,diencephalon, metencephalon, mesencephalon, myelencephalon
- ☒ .telencephalon, diencephalon, mesencephalon, metencephalon, myelencephalon
- ☐ .prosencephalon,diencephalon,mesencephalon,myelencephalon, metencephalon

76- The unspecialized germ cells found in the seminiferous tubules are called

- ☐ .Primary spermatocytes
- ☐ -Secondary spermatocytes
- ☒ -Spermatogonia
- ☐ .Primordial germ cells

77- Which process establishes the 3 definitive germ layers?

- ☐ . Neuralation
- ☒ . Gastrulation
- ☐ . Craniocaudal folding
- ☐ . Lateral folding

78- The average duration of gestation measured from the

first day of the last normal menstrual period

- ☐ 34 weeks
- ☐ .38 weeks
- ☒ .40 weeks
- ☐ .42 weeks

79- During week 2 the blastocyst implants and the outer cell layer in that week becomes:

- ☐ .decidual cells and hypoblast cell layer
- ☒ .cytotrophoblasts and syncytiotrophoblasts
- ☐ .chorion and amniotic membranes
- ☐ .the embryoblast and inner cell mass

80- Between which two layers is the extraembryonic mesoderm located?

- ☐ Epiblast and hypoblast
- ☐ . Syncytiotrophoblast and cytotrophoblast
- ☐ Syncytiotrophoblast and endometrium
- ☒ . Exocoelomic (Heuser's) membrane and cytotrophoblast

81- The intraembryonic coelom forms within:

- ☐ .somites
- ☒ .lateral plate
- ☐ .neural tube
- ☐ .intermediate mesoderm

82- The process of transformation of a spermatid into a spermatozoan is called as

- ☐ . Spermatogenesis
- ☐ .Gametogenesis
- ☒ .Spermiogenesis
- ☐ Sperm maturation

83- Pregnancy tests generally are based upon the detection in maternal urine of:

- . Estrogen
- . Progesterone
- . human Chorionic Somatomotropin
- ✓ . human Chorionic Gonadotrophin

Histology

1- Which cell is sensory reception between epidermal cells?

Lymphocyte

Melanocyte

Keratinocyte

✓ Merkel

2- Which type of junctions is an effective barrier to the diffusion of substances across an epithelium?

Gap junctions

Hemidesmosomes

✓ Tight junctions

Desmosomes

3- All of these are located in the chemical synapses, except:

Synaptic vesicles

Postsynaptic receptors

✓ Gap junctions

Mitochondria

4- Which layer of the epidermis has cells which have keratohyaline granules?

Stratum basale

✓ Stratum granulosum

Stratum corneum

Stratum spinosum

5- Which of the following is observed in the histological structure Metaphysis Plate

Osteoclast

Conderoclast

Osteocyte

✓ Chonderocyte

6- Which of muscle protein has three subunit(TnT,TnC,TnI)?

- ☒ Troponin
- Myosin
- Tropomyosin
- Actin

7- In which cell nucleus has cartwheel or clock face arrangement?

- Mastcell
- ☒ Plasmacell
- Fibroblast
- Fibrocyte

8- What is the covering of an individual muscle fiber?

- ☒ Endomysium
- Sarcolemma
- Perimysium
- Epimysium

9- What type of muscles have branching or spindle shape cells?

- Striated - Smooth
- ☒ Cardiac- Smooth
- Smooth- Striated
- Skeletal- Cardiac

10- Structure of framework of spleen and lymphnodes are composed of which fibers?

- elaunin
- ☒ reticular
- elastic
- collagen bundles

11- About Unmyelinated PNS axons, which choice is correct?

Oligodendrocyte cell starts to envelop one axon

Schwann cell starts to envelop one axon



Schwann cell starts to envelop multiple axons

Oligodendrocyte cell starts to envelop multiple axons

12- What do you call the simple squamous epithelium that lines the blood vessels?

Pseudostratified

Epithelioid tissue



Endothelium

Mesothelium

13- What is the following is INCORRECT?

cardiac muscle and skeletal muscle have sarcomere



Striation was seen in transverse section of cardiac muscle and skeletal muscle

cardiac muscle and smooth muscle nucleus located in center part

Striation was seen in longitudinal section of cardiac muscle and skeletal muscle

14- Which of the following histological structures has Chondroblast?

Wolkman Canal



Chondrium Matrix

Periosteum

Endosteum

15- Identify the INCORRECT statement

Epithelia are classified by the shape of cell layers

In pseudostratified epithelia all cells are in contact with the basement membrane.

Microvilli increase the surface area of the epithelia cells

The basement membrane consists of two sub layers: the reticular layer and the basal lamina.

16- A nerve is formed through a collection of

Myelin sheet



Axones

Glial cells

Neurons

17- What is the name of macrophage cells in liver?

Kupffer

Osteoclast

Langerhans

Microglial

18- Which cell has a part in formation of blood brain barrier?

Schwann cell

Oligodendrocyte

Ependymal cell



Astrocyte

19- In which cell granules composed of major basic proteins?

Neutrophil



Eosinophil

Basophil

Lymphocyte

20- Which layer of the epidermis contains cells with no nuclei or organelles?

granulosum

spinosum

basale



corneum

مدت آزمون: 75 بر اساس دفترچه کاربر Tissue, Development & Function: آزمون

21- An exocrine gland that loses cell bodies during secretion, as is the case for the sebaceous gland, is further classified as a.....gland.

- ☒ Holocrine
- ☐ Apocrine
- ☐ Endocrine
- ☐ Merocrine

22- This type of neuron consist of many dendrites and one true axon , and are more:

- ☒ Multipolar
- ☐ Unipolar or pseudo unipolar
- ☐ Bipolar
- ☐ Apolar

23- In which of the following is Observed only type I collagen ?

- ☒ Perichondrium
- ☐ Elastic Cartilage
- ☐ Hyalin Cartilage
- ☐ Matrix

24- Which cell is the largest white blood cell?

- ☐ Basophil
- ☐ Neutrophil
- ☐ Lymphocyte
- ☒ Monocyte

25- In which of the following is Observed Osteogenic cell &Osteoclast ?

- ☐ Periosteum
- ☐ Pericondrium
- ☐ Nothing
- ☒ Edosteum

26- Which cell secrete antibody in connective tissue?

Fibrocyte



Plasmacell

Mastcell

Adibocyte

27- What is line that bisects the dark band in muscle?

I band



M line

H band

A band

28- In which type of granules in platelets we can find lysosomic enzyme?

Landa

Delta

Beta

Alpha

29- Spinal central canal covered by:

Ependymal cell

Microglia

Oligodendrocyte

Schawann cell

30- Which cell has no nucleus?

Basophil

Normoblast



Erythrocyte

Neutrophil

31- Which of the following is a supporting cell found in the ganglia?

- ☒ Satellite cells
- ☐ Neuroglia
- ☐ Schwann cells
- ☐ Basket cells

32- In which tissue the nucleus are at the rim of the cells?

- ☐ Fibroblast
- ☐ Macrophage
- ☐ Fibrocyte
- ☒ Adipocyte

33- Nissl bodies in the perikaryon of neurons are composed of:

- ☒ Ribosomes and RER
- ☐ Golgi and RER
- ☐ Ribosomes and Lysosomes
- ☐ SER and Lysosomes

34- Which types of neurons are located in dorsal root ganglia?

- ☐ Multipolar
- ☐ Bipolar
- ☒ Pseudounipolar
- ☐ Anaxonic

35- What type of glands are the ceruminous glands in ear?

- ☐ Eccrine gland
- ☒ Apocrine gland
- ☐ Sebaceous glands
- ☐ Endocrine gland

36- Retrograde transportation in the axon is occurred by:

Talin

Kinesin

Actin

✓ Dynein

Embryology

37- Implantation of the blastocyst

cannot occur without prior

differentiation of syncytiotrophoblast

✓ .hatching from the zona pellucida

.fragmentation of the primary yolk sac

development of amnioblast

Embryology

38- A 4-year-old healthy female presents to your office to establish primary care. What phase of meiosis should her oocytes be in?

Meiosis I - prophase

✓

Meiosis II - prophase

Meiosis I - metaphase

Meiosis II – metaphase

39- Maternal serum alpha-fetoprotein increases if the fetus has

oligohydramnios

Bochdalek hernia

. Ladd's bands

✓ anencephaly

40- The intraembryonic coelom

- ✓ it is lined by endoderm
- normally contains only fluid
- becomes the pleural and peritonea
- , but not the pericardial, cavities
- all of the above

41- Pregnancy tests generally are based upon the detection in maternal urine of:

- Estrogen
- Progesterone
- human Chorionic Somatomotropin
- ✓ human Chorionic Gonadotrophin

42- The human placenta

- has a maternal component formed by the decidua capsularis
- shows no changes with age
- retains the cytotrophoblast layer of the placental barrier throughout gestation
- ✓ has chorionic villi as the structural and functional unit

43- As the follicle grows, the oogonium becomes located in a mound of follicle cells called the:

- vesicular antrum
- oocyte
- ✓ cumulus oophorus
- Mural granulosa cells

44- Neural crest

gives rise to most of the cardiogenic mesoderm



is derived from the epiblast

.becomes smooth muscle cells within the wall of the bowel

.induces development of syncytiotrophoblast

45- Which of the following is associated with oligohydramnios

tracheo-esophageal fistula



. renal agenesis

myelomeningocele

anencephal

46- A 3 year-old male patient presents with epicanthal folds, flat nasal bridge, single palmar crease. The patient is 47 XY on karyotype. What could be the cause of this karyotype?

Chromosomal mutation

. Chromosomal inversion



Chromosomal nondisjunction

Chromosomal deletion

47- Somites

are the source of cells in the nucleus pulposus

give rise to cardiac muscle

give rise to the cartilages of the branchial arches



are derived from the paraxial mesoderm

48- Implantation is preceded by an increase in ovarian hCG secretion

- ✓ usually occurs near the end of the first week following fertilization
- requires contact of the zona pellucida with uterine epithelium
- cannot occur outside the uterus and uterine tubes
- leads to the development of the inner cell mass

49- "Quickening" or the period when fetal movements are commonly first felt by the mother occurs:

- near the end of the first trimester
- ✓ around the middle of the second trimester
- at the end of the second trimester
- . at the end of the embryonic period

50- Extraembryonic somatic mesoderm is in close association with:

- syncytiotrophoblast
- ✓ cytotrophoblast & amnion
- yolk sac
- syncytio- & cytotrophoblast

51- The amniotic cavity develops:

- on the tenth day
- ✓ within the inner cell mass near the cytotrophoblast
- in extraembryonic mesoderm
- none of the above

52- The primitive streak

- ✓ is derived from epiblast
- first appears in the roof of the amniotic cavity
- gives rise to ectomesenchyme of the 1st branchial arch
- gives rise to the allantois

53- Major events beginning in the second week of development include:

notochord differentiation

somite formation

angiogenesis

✓ yolk sac development

54- 21 Mesoderm lateral to the notochord:

is derived from hypoblast

✓ differentiates into somite

migrates cranial to prochordal plate

induces endoderm differentiation

55- 22 Which of the following structures does not turn under onto the ventral surface of the embryo during folding of the head?

heart

✓ notochord

pericardial cavity

septum transversum

56- The outer layer of the umbilical cord is cont

inuous with, and similar in structure to the

chorionic membrane

✓ amniotic membrane

syncytiotrophoblast

cytotrophoblast

57- Which layer of the bilaminar embryonic disc forms part of the lining of the amniotic cavity?

epiblast



hypoblast

blastocyst

trophoblast

58- Myelomeningocele (using clinical terminology)

is most common at the tip of the coccyx



leaves the affected individual with paralyzed upper limbs but able to walk

is caused by a failure of the neural folds to meet and fuse

is undetectable by the results of amniocentesis

physiology

59- A motor unit consists of

A motor neuron and whole muscle it innervates.



A motor neuron and the group of skeletal muscle fibers it innervates.

An axonal terminal and whole muscle it innervates.

An axonal terminal and the group of skeletal muscle fibers it innervates.

60- Action potentials spread to the interior of the skeletal muscle fiber by way of

terminal cisternae



transverse tubules

endoplasmic reticulum

longitudinal tubules

مدت آزمون: 75 بر اساس دفترچه کاربر Tissue, Development & Function: آزمون

61- Compared to the skeletal muscle fibers, what is the cause of slow cycling of the myosin cross-bridges in the smooth muscle cells?

Low affinity to Ca ions in smooth muscle fibers



Far less ATPase activity of smooth muscle fibers

Higher activity of myosine kinase

Response to different types of stimuli by smooth muscle fibers

62- In which of the following muscle, control is exerted mainly by nerve signals?

Uterus

Ureters

Bile Duct



Ciliary Muscle

63- Which of the following best refers to the importance of chemical synapses?

Synchronous firing of interconnected cells

Rapid firing



Signals amplification

fast conduction velocity

64- During rigor mortis, contraction cycle is stopped at step 3 because of



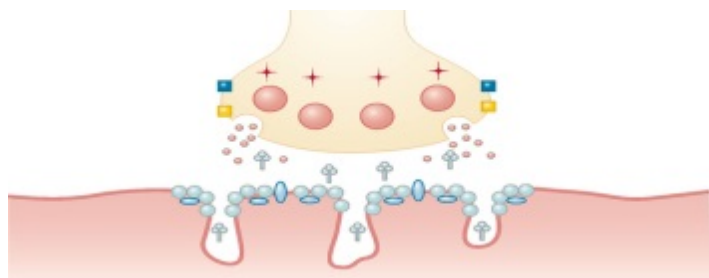
lack of ATP

low affinity of actin filaments to myosin heads

non-functional heads of myosin filaments

ADP and Pi release

65-  In this figure, what does this sign shows?



Acetylcholine molecules

Dens bar

Ca ions



Acetylcholine esterase

66- Which of the following best describes the significant difference between skeletal muscle and large nerve fibers?

Generation of action potential

Voltage gated sodium channels presense

The amount of resting membrane potential



Velocity of conduction

67- Which of the following represents one of the electrical synapses properties?

Presynaptic vesicles

Chemical transmitter

Synaptic delay



Bidirectional transmission

68- Which of the following is correct about isometric contraction?

The muscle shortens.

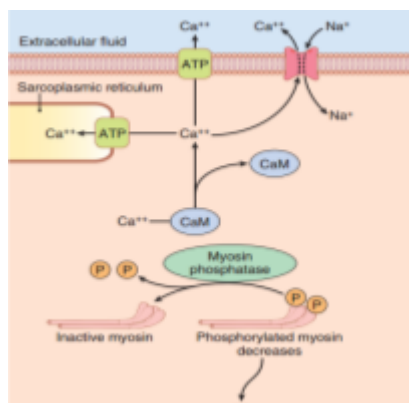
The tension on the muscle remains constant during the contraction.



The load can not be lifted.

It occurs when the force of the muscle contraction is greater than the load.

69-



What does this figure show?

Stress relaxation phenomenon

Smooth muscle contraction

Latch mechanism

Smooth muscle relaxation



70- Which of the following is believed to initiate the contraction process in the skeletal muscle fibers?

Sliding filament mechanism

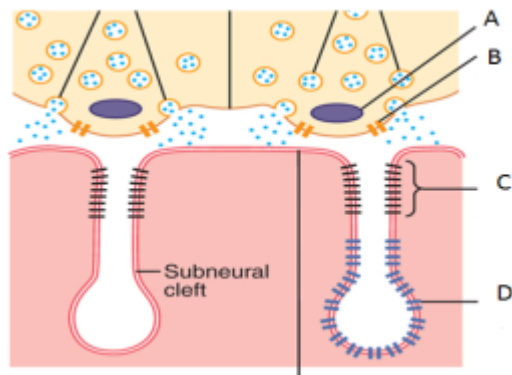
Adenosine triphosphatase activity of the myosin head

Actin-myosin attachment

The strong affinity of the troponin C for calcium ions



71- Which of the following represents "Acetylcholine gated channels"?



A

✓ C

D

B

72- Which of the following is not matched the excitation contraction coupling in the skeletal muscles?

Dihydropyridine receptors

Transverse tubules

Ryanodine receptors

✓ Multiple adherent points

73- In an electromyography report, findings are compatible with severe axonal lesion of a nerve supply to a skeletal muscle. What symptom do you expect to observe in this patient?

Hyperplasia of muscle fibers

✓ Atrophic changes

Remodeling of muscle fibers

Muscle fatigue

74- Which factor inhibits the active sites of actin filaments?

- Myosin heads detachment
- ATP binding to actin filaments
- Inactivation of myosin filaments
- ✓ Tropomyosin-troponin complex

75- In this type of muscle, the source of ca ions needed for contraction is just from sarcoplasmic reticulum?

- Blood vessels
- ✓ Biceps muscle
- Uterus
- Iris muscle of the eye